

LA-ICPMS ANALYSIS OF THE PREDOMINANT ORE MINERALS FROM VARIOUS TYPES OF ORES WITHIN THE OKTYABRSKOE DEPOSIT (TALNAKH-NORIL'SK ORE DISTRICT)

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There are unique Cu-Ni-Pt-Pd deposits associated with trap magmatism arising at the P-T boundary in the north-eastern part of the Siberian Platform (Ryabov et al., 2014). The main feature of these deposits is the predominance of Pd over Pt, which leads to 75% of world production of Pd. One of the largest deposits in this region is the Oktyabrskoe deposit. It is located and genetically related to the Kharaelakh intrusion. The host rocks for the deposit are terrigenous aluminosilicate and carbonate-sulfate sediments of Devonian (Zolotukhin et al., 1975).

The predominant sulfides within the Oktyabrskoe deposit are the minerals of the pyrrhotite group, chalcopyrite, and pentlandite, and also in significant amounts are noted such minerals as cubanite, talnakhite, mooihoekite, millerite, bornite, valleriite, covellite, heazlewoodite and others. Sulfide minerals are part of the industrial types of ores, which are subtracted by their structural and mineralogical characteristics. The major industrial types are massive (rich), disseminated and lenticular-disseminated (sometimes called «brecciated» or «copper-bearing»).

In massive ores, the sulfide mineralization reaches to 85-95%, they are usually concentrated in the intrusion base, on contact with host rocks, sometimes forming fissile veins over the host sedimentary rocks. Lenticular-disseminated ores contain from 30 to 50% sulfides. They are concentrated on the front face and in the roofing parts of the intrusion. Disseminated ores are located in the intrusion body, in the picritic and taxitic gabbro-dolerites horizon. The content of sulfide minerals varies from the first percent to 15-20%.

According to classical concepts, the formation of the massive sulfide ores is associated with the processes of liquation between sulfide and aluminosilicate melts. When the sulfide liquid was separated from the aluminosilicate part by the gravitational forces, the sulfide liquid sank out to the bottom of the magmatic chamber, forming a horizon of massive sulfide ores. At the same time, some drops that did not have time to sink out to the bottom of the chamber crystallized in the intrusion body, forming a zone of disseminated ores. Lenticular-disseminated ores are associated with contact-metamorphic and metasomatic mineral associations.

We have studied 3 main sulfide minerals (pyrrhotite, chalcopyrite, pentlandite) from three samples of various industrial types of ore. These specimens were studied under scanning electron microscope (SEM) TESCAN VEGA 3 SBU and energy-dispersive adapter OXFORD X-Max 50 with 20 kV accelerating voltage, sample current 10-50 nA (TPU, Tomsk). An example of SEM visualization and analyzes is shown in Fig. 1a,b. The composition of trace element was determined by inductively coupled plasma mass spectrometry with laser ablation (LA-ICPMS) (IGEM, Moscow). The instrument was Thermo Xseries 2 quadrupole mass spectrometer with ESI NWR 213 laser. The beam size was 20-30 µm (Fig. 1a). Three sulfide reference materials were used to control the quality of analytical results: MASS-1 (US Geological Survey) - certified Synthetic Polymetallic Sulfide ZnCuFeS as pressed pellets; in-house standard po-stc, containing 20 ppm Au, Ag and PGE in a pyrrhotite matrix, manufactured according to a method described in (Wohlgenuth-Ueberwasser et al., 2007); UQAC-FeS1 (UQAC, Chicoutimi, Canada) pressed natural sulfide doped with trace elements. Iron isotope-57 was used as an internal standard.

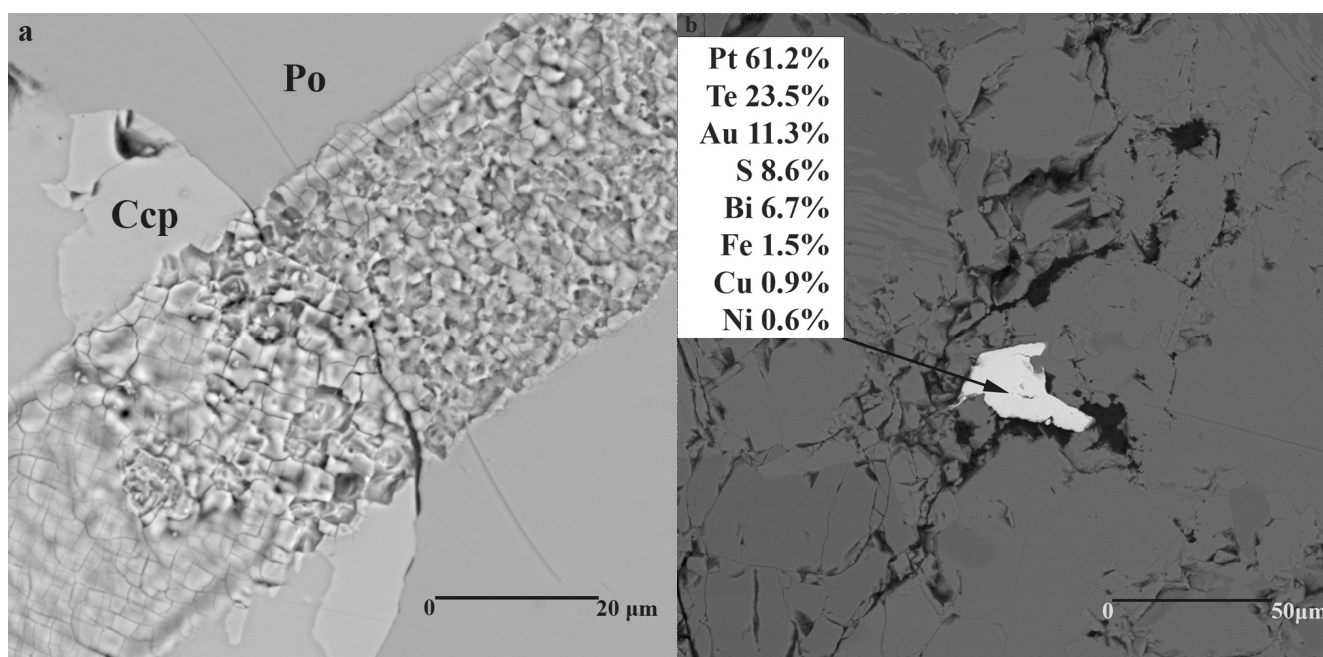


Fig. 1. An example of SEM visualization. In Fig.1a a laser ablation track is evident across chalcopyrite (Ccp) and pyrrhotite (Po) boundary. In Fig. 1b non-identified mineral (Pd-Au-rich cooperite?) is shown.

Sample AM59 is a massive ore of chalcopyrite-pyrrhotite, containing a few insets of such minerals as anhydrite, calcite, and chlorite. Ore mineralization is represented by pyrrhotite, chalcopyrite, pentlandite, and Co-rich pentlandite. Cooperite, Pd-Au-Pt sulfide (Pd-Au-rich cooperite? (Fig. 1.b)) and native gold with an admixture of silver are also present. PGE minerals of 25-50 μm in size are usually confined to the boundaries of sulfide phases.

Sample AM51 represents lenticular-disseminated ore. Sulfide mineralization is represented by pyrrhotite, chalcopyrite, pentlandite, Ag-, Co-containing pentlandite. The silicate part is represented by the hedenbergite-diopside pyroxene, the composition of pyroxene varies, from virtually pure diopside to pure hedenbergite. According to the results of LA-ICPMS, pyrrhotite is characterized by traces of Co (90 ppm), Ni (0.2%), Pb (14 ppm), chalcopyrite - Co (4-10 ppm), Ag (1-10 ppm), Cd (30-600 ppm), In (2-5 ppm), Te (10-35 ppm), Pb (5-40 ppm), pentlandite - Co (0.8-1.2%), Pd (470-850 ppm), Ag (0.2-2.5 ppm).

Sample AM68 represents disseminated ore. Sulfide mineralization is represented by unevenly distributed sulfide, amoebae drops. The composition of the drops is pyrrhotite, chalcopyrite, pentlandite, Co-rich pentlandite. According to the results of LA-ICPMS analysis, pyrrhotite contains Ni (1-1.5%), Co (200-770 ppm), Cd (46-63 ppm), In (0.3 ppm), Pb (0.7-20 ppm), Bi (0.2- 5 ppm), chalcopyrite - Co (0.8-45 ppm), Ni (50-75 ppm), Ag (3-30 ppm), Cd (35-100 ppm), Sn (16-25 ppm), Pb (2.3- 10 ppm) Bi (0.6-2 ppm). Pentlandite is rare and was not sampled by the laser ablation tracks.

The most characteristic trace elements in sulfide minerals are shown in Table 1. It can be seen that Ni, Co, Pb are permanent element in pyrrhotite. Pt was found only in pyrrhotite from massive ore, while pyrrhotite of disseminated ore is characterized by the presence of Cd, In, Bi, and pyrrhotine from lenticular-disseminated ore is the poorest in trace elements.

Table 1. The most common trace elements in major sulfide minerals are industrial types of ores.

Minerals	pyrrhotite	chalcopyrite	pentlandite
Ore types			
Massive	Ni, Co, Pb, Pt	Ag, Cd, Pt, Pb, Bi	Co, Pd, Au
Lenticular-disseminated	Ni, Co, Pb	Co, Ag, Cd, Pb, In, Te,	Co, Pd, Ag
Disseminated	Ni, Co, Pb, Cd, In, Bi	Ni, Co, Ag, Cd, Pb, Bi, Sn	Not analysed

Ag, Cd and Pb impurities are characteristic of chalcopyrite from all types of ores. Perhaps this is due to the inclusion of galena. Pt was found only in chalcopyrite from massive ore. The analyzed pentlandite is characterized by a constant admixture of Co and Pd, while Au and Ag are found in varying amounts.

Abnormally high concentrations of Pd are found only in pentlandite. The analysis was performed at surface areas with no visible inclusions of other minerals. However, it is possible that tiny inclusions of the Pd minerals were located below the surface and were captured during the analysis. While for Ag and Au, it is possible to assume their isomorphic entry into the lattice of sulfide minerals.

Thus, there are some patterns in the distribution of trace elements in the main sulfides of the three types of ores:

1. Pt is a characteristic of pyrrhotite and chalcopyrite of massive ores only.
2. Chalcopyrite and pyrrhotite of disseminated ores are characterized by the greatest variety of trace elements.
3. Explicit fractionation of Pt and Pd is detected. Pd is concentrated in pentlandite only.

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